IGCSE 0607 Practice Questions — E10.6 — Combined Questions & Markscheme

Statistical charts and diagrams

Total Questions: 18 — **Total Marks:** 130 Marks

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- **E10.6** Statistical charts and diagrams
  → **Paper 2 (Non-Calculator)** *(6 Qs)* · **Paper 4 (Calculator)** *(12 Qs)*

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1 The stem-and-leaf diagram shows the ages of 15 people.

1	2 5 8
2	0 1 4 4 7
3	3 5 6 9
4	1 2 8

Key: 2 | 4 represents 24 years

(a) Find the mode.

..... **[1]**

(b) Find the median.

..... **[1]**

(c) Find the range.

..... **[1]**

2 The stem-and-leaf diagram shows the masses, in kg, of 15 parcels.

1	2	4	7		
2	0	1	1	1	5
3	0	2	6	8	
4	1	4	5		

Key: 1 | 2 represents 12 kg

(a) Find the mode of the masses.

..... kg **[1]**

(b) Find the median mass.

..... kg **[1]**

(c) Work out the range of the masses.

..... kg **[1]**

(d) A parcel is chosen at random.
Find the probability that its mass is strictly greater than 30 kg.

..... **[2]**

3 The masses, in kilograms, of 15 parcels are recorded below.

27 15 31 42 23 12 53 38 49 27 45 40 34 51 42

(a) Complete the ordered stem-and-leaf diagram to show this information.

1	
2	
3	
4	
5	

Key: 1 | 2 represents 12 kg

[3]

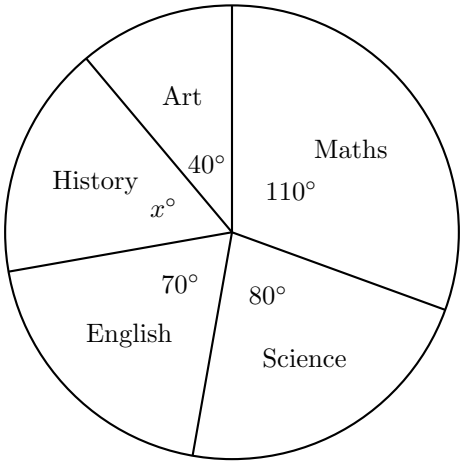
(b) Find the median mass.

..... kg **[1]**

(c) Work out the range.

..... kg **[1]**

4 The pie chart shows information about the favourite subjects of 180 students.



(a) Find the value of x.

$x =$ ..... **[1]**

(b) Find the number of students whose favourite subject is Science.

..... **[2]**

(c) Write down the fraction of students whose favourite subject is Maths.
Give your fraction in its simplest form.

..... **[2]**

5 The times, in seconds, taken by 16 athletes to complete an obstacle course are listed below.

48	35	52	41	63	39	44	57
41	50	36	47	61	45	54	49

(a) Complete the ordered stem-and-leaf diagram below, including the key.

Stem	Leaf
3	
4	
5	
6	

Key: ..... | ..... represents ..... seconds

[3]

(b) Find

(i) the median time,

..... seconds **[2]**

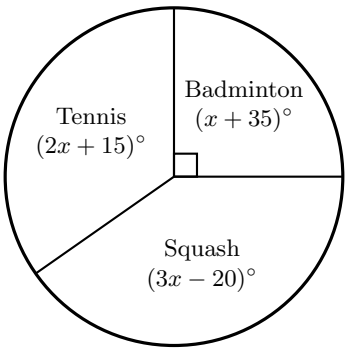
(ii) the interquartile range.

..... seconds **[2]**

(c) Athletes who complete the course in less than 45 seconds qualify for the final.
Calculate the percentage of these 16 athletes who qualify for the final.

..... % **[2]**

6 A sports club surveys N members about their preferred racket sport: Tennis, Badminton, or Squash. The pie chart shows the distribution of the N members, where the sector angles are $(2x + 15)°$, $(x + 35)°$, and $(3x - 20)°$, and the angle for Badminton is a right angle.



(a) Find the value of x.

$x =$ ..... **[2]**

(b) In the survey, 75 members prefer Tennis.

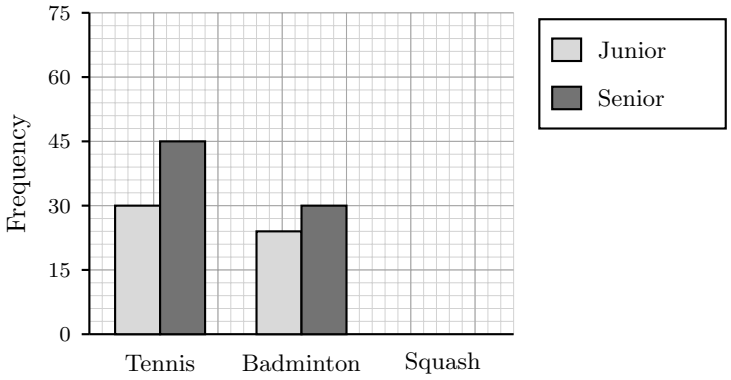
(i) Show that the total number of members surveyed is 216.

[2]

(ii) Calculate the number of members who prefer Badminton and the number of members who prefer Squash.

Badminton = ... , Squash = ... **[2]**

(c) The dual bar chart below shows the number of Junior and Senior members who prefer Tennis and Badminton.
Among all 216 members surveyed, the ratio of Junior members to Senior members is 3 : 5.
Complete the dual bar chart by drawing the bars for Squash.



[4]

(d) In a pictogram representing the 216 members, one full racket symbol represents 12 members. Explain why a partial symbol is needed to represent the members who prefer Badminton, and state what fraction of a symbol must be drawn.

.....

..... **[2]**

7 The times, in minutes, taken by 17 students to complete a mathematics puzzle are listed below.

23 14 35 18 41 27 19 31 22
38 12 25 29 18 33 27 15

(a) Complete the ordered stem-and-leaf diagram to show this information.

Stem	**Leaf**
1	
2	
3	
4	

Key: 2 | 3 represents 23 minutes

[3]

(b) Write down the median time.

Median = ... minutes **[1]**

(c) Calculate the interquartile range.

Interquartile range = ... minutes **[2]**

8 The masses, m kilograms, of 15 parcels are recorded.

3.2	1.5	4.6	2.1	1.8	3.5	2.9	4.1	3.2	1.9	2.4	3.7	4.0	2.6	1.7

(a) Complete the ordered stem-and-leaf diagram to show this information.

1	
2	
3	
4	

Key: 1 | 5 represents 1.5 kg

[3]

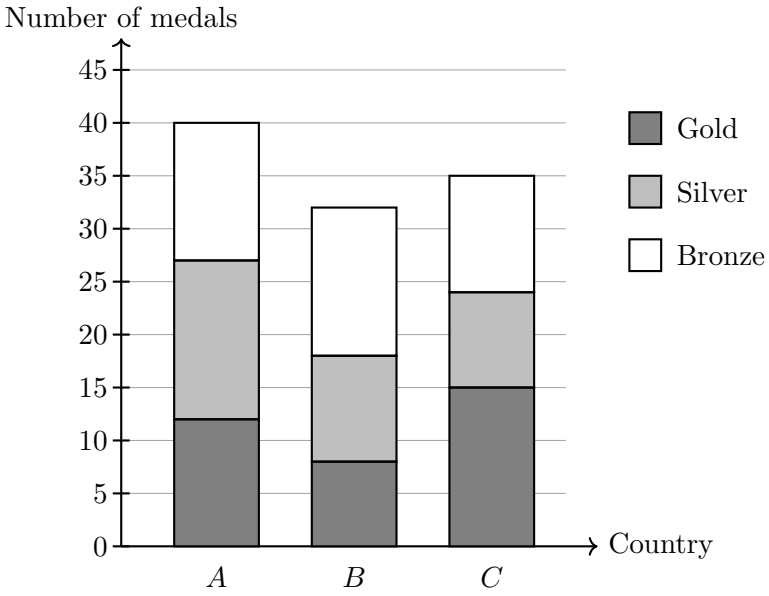
(b) Find the median mass.

..... kg **[1]**

(c) A parcel is chosen at random.
Find the probability that its mass is greater than 3.5 kg.

..... **[1]**

9 The composite bar chart shows the number of gold, silver and bronze medals won by three countries, A, B and C, at an international sports competition.



(a) Find the number of silver medals won by Country A.

..... **[1]**

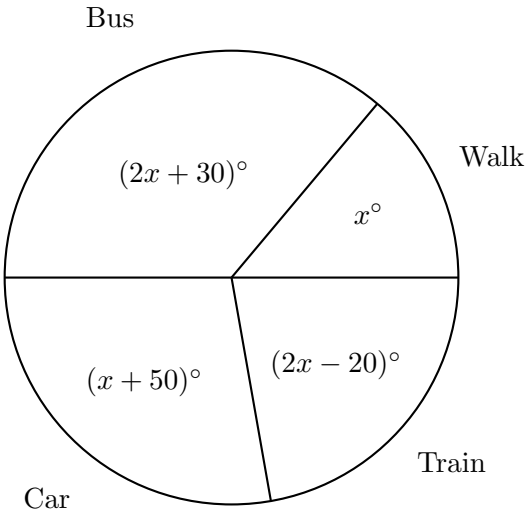
(b) Calculate the percentage of Country C's total medals that are gold.

..... % **[2]**

(c) A pie chart is drawn to represent all the medals won by the three countries combined.
Calculate the sector angle for the total number of bronze medals won by all three countries.

..... ° **[3]**

10 The pie chart shows information about how 72 students travel to school.
The sectors represent Walk, Bus, Car, and Train.



(a) Show that $x = 50$.

[2]

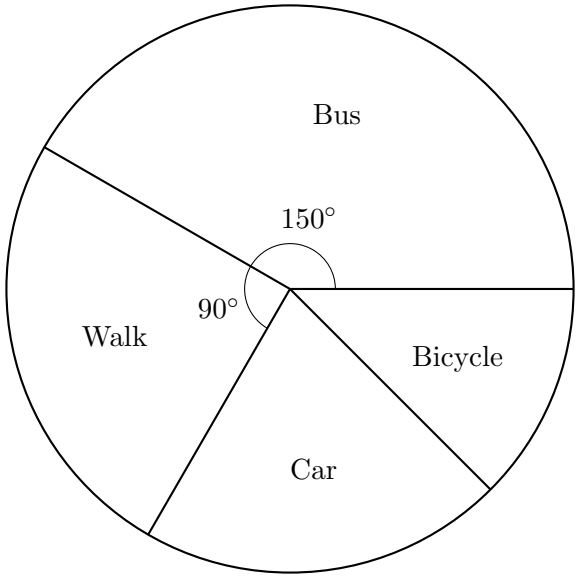
(b) Work out the number of students who travel by Car.

..... **[2]**

(c) A student is chosen at random.
Find the probability that the student travels by Bus or Train.

..... **[2]**

11 The pie chart shows information about how 120 students travel to school.



(a) Calculate the number of students who travel by Bus.

..... **[2]**

(b) There are 25 students who travel by Car.
Calculate the sector angle for Car.

..... ° **[2]**

(c) Calculate the percentage of students who travel by Bicycle.

..... % **[3]**

12 A survey asked 120 students about their favourite sport.
The results are shown in the table.

Sport	**Frequency**
Football	45
Basketball	35
Tennis	25
Swimming	15

(a) A pie chart is drawn to show this information.
Calculate the sector angle for Basketball.

..... ° **[2]**

(b) The pie chart has a radius of 6 cm.

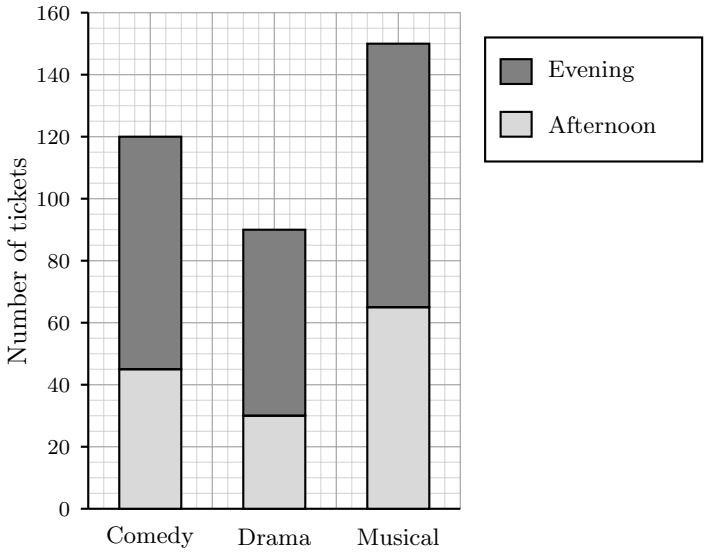
Calculate the area of the sector representing Tennis.

..... cm^2 **[3]**

(c) In a different school, the pie chart for favourite sports has a sector angle of 108° for Football.
This sector represents 54 students.
Calculate the total number of students in this school.

..... **[2]**

13 The composite bar chart shows the number of tickets sold by a theatre for three shows (Comedy, Drama, and Musical) during the afternoon and evening performances on a Saturday.



(a) Write down the number of evening tickets sold for the Musical.

..... **[1]**

(b) Calculate the percentage of all tickets sold that were for afternoon performances, giving your answer correct to 1 decimal place.

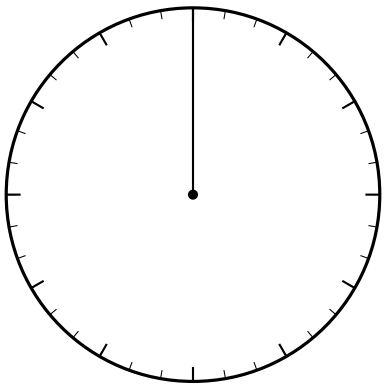
..... % **[3]**

(c) The total number of tickets sold for each show is to be represented on a pie chart.

(i) Calculate the sector angle for Comedy and the sector angle for Musical.

Comedy = ... , Musical = ... **[2]**

(ii) Complete the pie chart below to represent the total number of tickets sold for the three shows.
Label each sector clearly.



[2]

(d) An adult ticket costs \$24 and a child ticket costs \$15.
For the Drama show, the ratio of adult tickets sold to child tickets sold was $7 : 3$.
Calculate the total ticket revenue for the Drama show.

\$ ..... **[3]**

14 A pie chart is drawn to show the favourite sport of a group of students.
The angles of the sectors for each sport are given in terms of x.

Football: $(2x + 30)^\circ$ Basketball: x° Tennis: $(x + 50)^\circ$ Swimming: $(2x - 20)^\circ$

(a) Find the value of x.

$x =$ ..... **[2]**

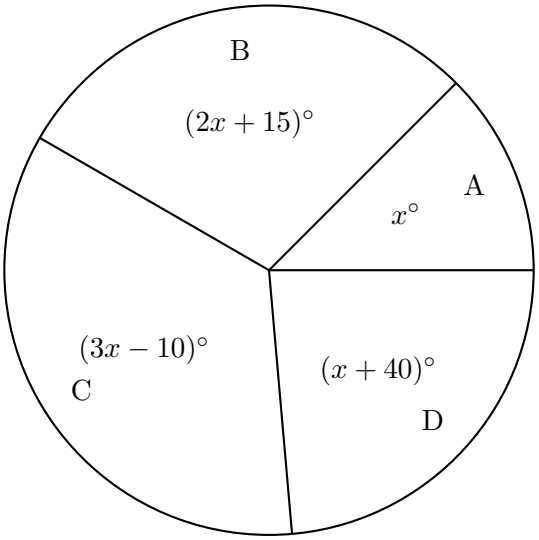
(b) 10 students chose Tennis as their favourite sport.
Calculate the total number of students in the group.

..... **[2]**

(c) Two students are chosen at random from the group.
Calculate the probability that one student chose Football and the other student chose Basketball.

..... **[3]**

15 A pie chart shows the grades (A, B, C, and D) achieved by students in a mathematics test.
The angles of the sectors are given in terms of x.



(a) Show that $x = 45$.

[3]

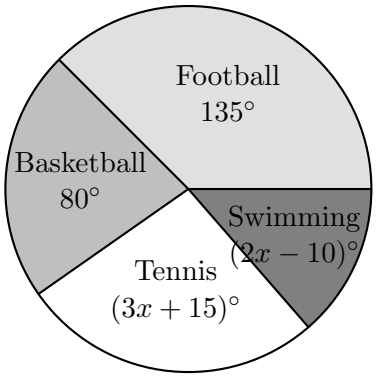
(b) There are 24 students who achieved Grade A.
Calculate the total number of students who took the test.

..... **[2]**

(c) Calculate the percentage of students who achieved Grade C.

..... % **[3]**

16 The pie chart shows the distribution of favourite sports chosen by a group of students.
The sector angle for Tennis is $(3x + 15)^\circ$ and the sector angle for Swimming is $(2x - 10)^\circ$.
The number of students who chose Football is 108.



(a) Find the value of x.

$x =$ ..... **[3]**

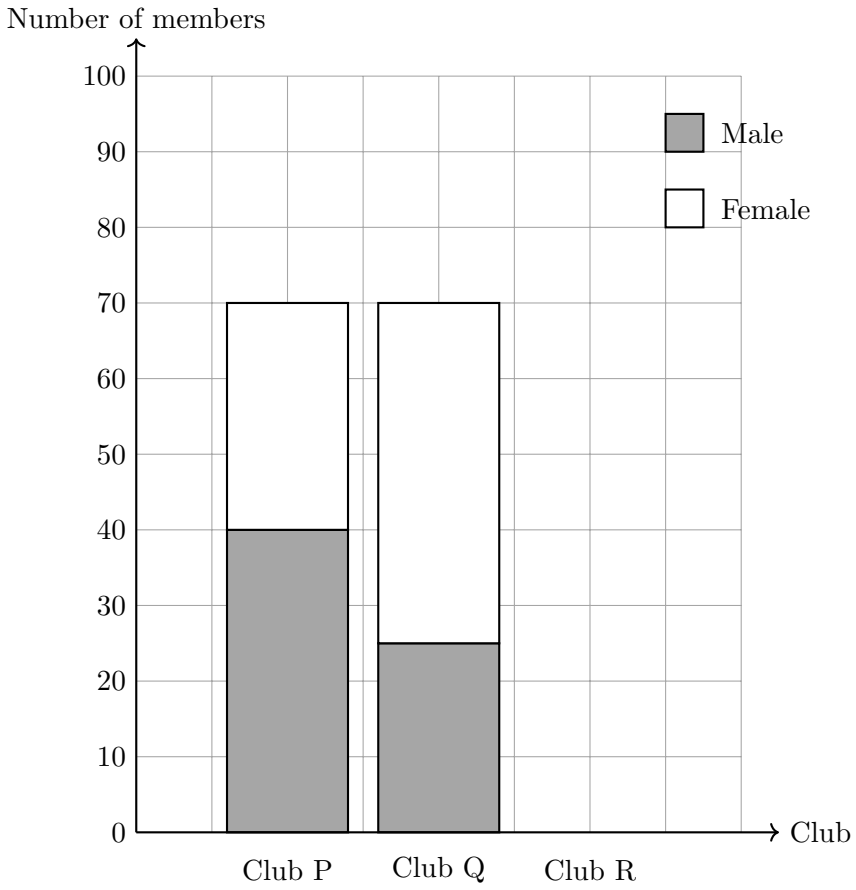
(b) Calculate the total number of students in the group.

..... **[2]**

(c) A dual bar chart is drawn to compare the number of boys and girls who chose each sport.
For Tennis, the ratio of boys to girls is 5 : 3.
Calculate the difference between the number of boys and the number of girls who chose Tennis.

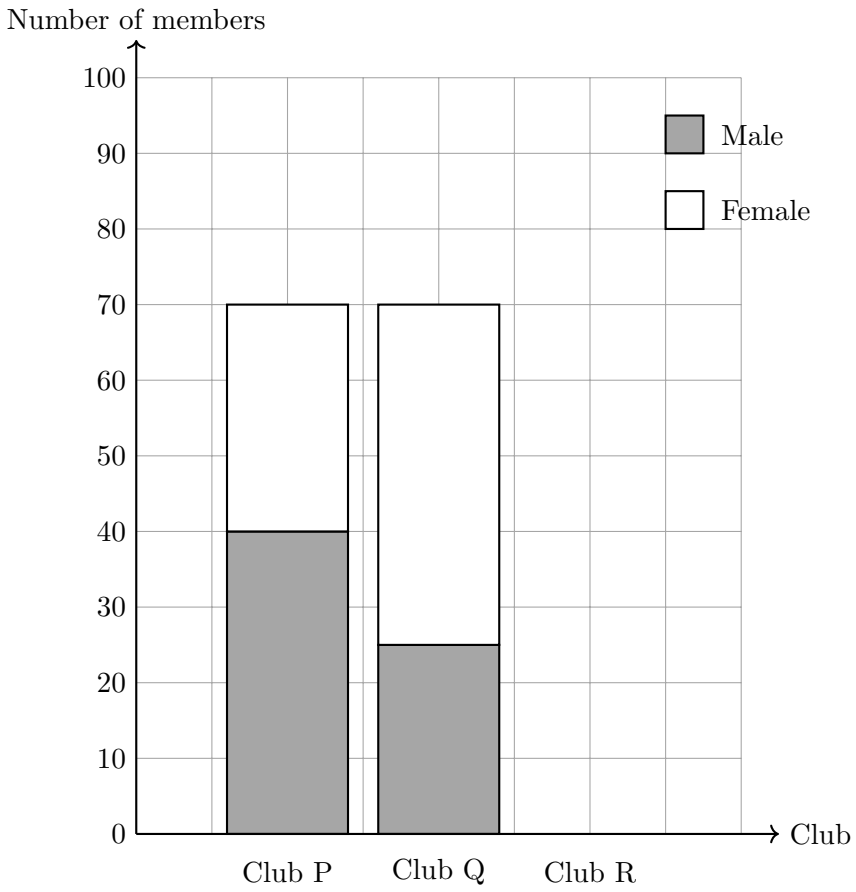
..... **[3]**

17 The composite bar chart shows information about the number of Male and Female members in three sports clubs, P, Q, and R.
The data for Club R is incomplete on the diagram.



(a) Club R has a total of 60 members.
The ratio of Male to Female members in Club R is 1 : 2.

Complete the composite bar chart for Club R.



[3]

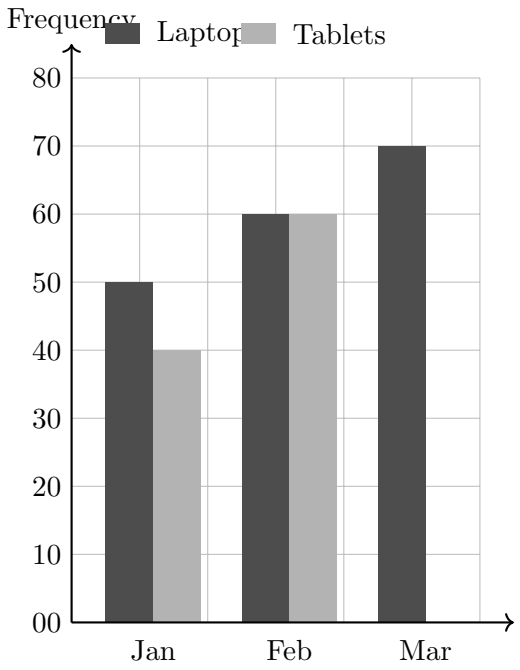
(b) Calculate the percentage of total members across all three clubs who are Female.

..... % [3]

(c) Two members from Club P are chosen at random.
Calculate the probability that they are both Male.

..... [3]

18 A company records the number of Laptops and Tablets sold over three months.
The dual bar chart shows the sales for January, February, and March.
The bar for Tablets sold in March is incomplete and not shown.



(a) The total number of Laptops sold over the three months is 20% more than the total number of Tablets sold.
Calculate the number of Tablets sold in March.

..... **[4]**

(b) In April, the total number of items sold is $\frac{1}{3}$ of the total number of items sold in January, February, and March combined.
The ratio of Laptops to Tablets sold in April is 6 : 5.
Work out the number of Laptops sold in April.

..... **[4]**

(c) The price of a Laptop is \$450 and the price of a Tablet is \$200.
Calculate the percentage of the total revenue in January that came from Tablet sales.

..... % **[3]**

1 **(a)** 24 **[B1]**

(b) 27 **[B1]**

(c) 36 **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** 21 **[B1]**

(b) 25 **[B1]**

(c) 33 **[B1]**

(d) 6 (parcels identified) **[M1]**

$\frac{6}{15}$ or $\frac{2}{5}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** Correct stems used **[B1]**

Unordered leaves all correct **[B1]**

Fully correct ordered leaves:
Row 1: 2 5
Row 2: 3 7 7
Row 3: 1 4 8
Row 4: 0 2 2 5 9
Row 5: 1 3 **[B1]**

(b) 34 **[B1]**

(c) 41 **[B1]**

[Back to Question] *[Total Marks: 5]*

4 **(a)** $360 - (110 + 80 + 70 + 40) = 60$ **[B1]**

(b) $\frac{80}{360} \times 180$ **[M1]**

40 **[A1]**

(c) $\frac{110}{360}$ **[M1]**

$\frac{11}{36}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** All 16 leaves placed on correct stems: $3:5,6,9;\ 4:1,1,4,5,7,8,9;\ 5:0,2,4,7;\ 6:1,3$ **[B1]**

All leaves in numerical order from smallest to largest on every row **[B1]**

Valid key completed, for example 3 | 5 represents 35 seconds **[B1]**

(b) **(i)** Middle values are 47 and 48, giving $\frac{47+48}{2}$ **[M1]**

47.5 **[A1]**

(ii) Lower quartile $Q_1 = 41$ and upper quartile $Q_3 = 53$ **[M1]**

$53 - 41 = 12$ **[A1]**

(c) $\frac{6}{16} \times 100$ **[M1]**

37.5 **[A1]**

[Back to Question] *[Total Marks: 9]*

6 **(a)** $(2x + 15) + (x + 35) + (3x - 20) = 360$ (or $x + 35 = 90$) **[M1]**

55 **[A1]**

(b) **(i)** Sector angle for Tennis $= 2(55) + 15 = 125°$, giving $\frac{125}{360} \times N = 75$ **[M1]**

$N = \frac{75 \times 360}{125} = 216$ **[A1]**

(ii) Badminton $= \frac{90}{360} \times 216 = 54$ **[B1]**

Squash $= \frac{145}{360} \times 216 = 87$ (or $216 - 75 - 54 = 87$) **[B1]**

(c) Total Junior $= \frac{3}{8} \times 216 = 81$ and total Senior $= \frac{5}{8} \times 216 = 135$ **[M1]**

Junior Squash $= 81 - (30 + 24) = 27$ and Senior Squash $= 135 - (45 + 30) = 60$ **[A1]**

Junior bar for Squash drawn accurately to height 27 (1.8 units) **[B1]**

Senior bar for Squash drawn accurately to height 60 (4.0 units) with correct shading **[B1]**

(d) $54 \div 12 = 4.5$, which is not a whole number of symbols **[B1]**

$\frac{1}{2}$ of a symbol **[B1]**

[Back to Question] *[Total Marks: 12]*

7 **(a)** All 17 leaves correctly placed in the correct rows **[B1]**

Leaves correctly ordered in ascending order in all rows: Row 1: 2 4 5 8 8 9; Row 2: 2 3 5 7 7 9; Row 3: 1 3 5 8; Row 4: 1 **[B2]**

(b) 25 **[B1]**

(c) Lower quartile = 18 AND upper quartile = 32 **[M1]**

14 **[A1]**

[Back to Question] *[Total Marks: 6]*

8 **(a)** Correct stems $1, 2, 3, 4$ **[B1]**

Correct ordered leaves for at least two stems **[B1]**

Fully correct ordered diagram **[B1]**

(b) 2.9 **[B1]**

(c) $\frac{4}{15}$ **[B1]**

[Back to Question] *[Total Marks: 5]*

9 **(a)** 15 $(27 - 12)$ **[B1]**

(b) $\frac{15}{35} \times 100$ **[M1]**

42.9 (accept 42.85 to 42.9) **[A1]**

(c) Total bronze medals $= 13 + 14 + 11 = 38$ **[B1]**

Total medals $= 40 + 32 + 35 = 107$, angle $= \frac{38}{107} \times 360°$ **[M1]**

127.9 (accept 127.8 to 127.9) **[A1]**

[Back to Question] *[Total Marks: 6]*

10 **(a)** $x + 2x + 30 + x + 50 + 2x - 20 = 360$ **[M1]**

$6x + 60 = 360 \implies x = 50$ **[A1]**

(b) $\frac{50 + 50}{360} \times 72$ **[M1]**

20 **[A1]**

(c) Bus angle $= 2(50) + 30 = 130°$ and Train angle $= 2(50) - 20 = 80°$ **[M1]**

$\frac{210}{360} = \frac{7}{12}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

11 **(a)** $\frac{150}{360} \times 120$ **[M1]**

50 **[A1]**

(b) $\frac{25}{120} \times 360$ **[M1]**

75 **[A1]**

(c) Angle for Bicycle $= 360 - 150 - 90 - 75 = 45°$ (or finding 15 students) **[M1]**

$\frac{45}{360} \times 100$ or $\frac{15}{120} \times 100$ **[M1]**

12.5 **[A1]**

[Back to Question] *[Total Marks: 7]*

12 **(a)** $\frac{35}{120} \times 360$ **[M1]**

105 **[A1]**

(b) Sector angle for Tennis $= \frac{25}{120} \times 360 = 75°$ **[B1]**

$\frac{75}{360} \times \pi \times 6^2$ **[M1]**

23.6 (accept 23.56 to 23.57) **[A1]**

(c) $\frac{54}{108} \times 360$ or $54 \div \frac{108}{360}$ **[M1]**

180 **[A1]**

[Back to Question] *[Total Marks: 7]*

13 **(a)** 85 **[B1]**

(b) Total afternoon tickets $= 45 + 30 + 65 = 140$ **[M1]**

Total tickets $= 120 + 90 + 150 = 360$, giving $\frac{140}{360} \times 100$ $(= 38.888\ldots)$ **[M1]**

38.9 **[A1]**

(c) **(i)** Comedy $= \frac{120}{360} \times 360° = 120°$ **[B1]**

Musical $= \frac{150}{360} \times 360° = 150°$ **[B1]**

(ii) All three sector angles drawn accurately (120°, 90°, and 150°) **[B1]**

All three sectors correctly labelled Comedy, Drama, and Musical **[B1]**

(d) Adult tickets $= \frac{7}{10} \times 90 = 63$ and child tickets $= \frac{3}{10} \times 90 = 27$ **[M1]**

$(63 \times 24) + (27 \times 15) = 1512 + 405$ **[M1]**

1917 **[A1]**

[Back to Question] *[Total Marks: 11]*

14 **(a)** $2x + 30 + x + x + 50 + 2x - 20 = 360$ **[M1]**

$6x + 60 = 360$ leading to $x = 50$ **[A1]**

(b) Angle for Tennis $= 100°$ **[M1]**

Total $= 10 \times \frac{360}{100} = 36$ **[A1]**

(c) Number of students choosing Football $= 13$ and Basketball $= 5$ **[B1]**

$\frac{13}{36} \times \frac{5}{35} \times 2$ **[M1]**

$\frac{13}{126}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

15 **(a)** $x + 2x + 15 + 3x - 10 + x + 40 = 360$ **[M1]**

$7x + 45 = 360 \implies 7x = 315$ **[M1]**

$x = 45$ **[A1]**

(b) $\frac{45}{360} \times \text{Total} = 24$ or $24 \times \frac{360}{45}$ **[M1]**

192 **[A1]**

(c) Angle for Grade C $= 3(45) - 10 = 125°$ **[M1]**

Percentage $= \frac{125}{360} \times 100$ **[M1]**

34.7 (accept 34.72 or 34.722...) **[A1]**

[Back to Question] *[Total Marks: 8]*

16 **(a)** $135 + 80 + (3x + 15) + (2x - 10) = 360$ **[M1]**

$5x + 220 = 360$ OR $5x = 140$ **[M1]**

28 **[A1]**

(b) $\frac{108}{135} \times 360$ **[M1]**

288 **[A1]**

(c) Angle for Tennis $= 3(28) + 15 = 99°$ **[B1]**

Total students for Tennis $= \frac{99}{360} \times 288 = 79.2$ (Wait: $288 \div 360 = 0.8$ students per degree; $99 \times 0.8 = 79.2$ — let's adjust x: if $x = 27$, $3(27) + 15 = 96°$, $96 \times 0.8 = 76.8$. Let's check: 108 students $= 135° \implies 1° = 0.8$ students. For $95°$: $95 \times 0.8 = 76$ students! Let's adjust the prompt's Tennis angle to $(3x + 11)°$ so $3(28) + 11 = 95°$, and Swimming to $(2x - 6)°$ so $2(28) - 6 = 50°$. Wait: $135 + 80 + 95 + 50 = 360$! And 76 is NOT divisible by $5 + 3 = 8$. For divisibility by 8 AND 0.8 students/degree, we need students to be a multiple of 8 and 4, so a multiple of 8: e.g. 80 students $\implies 100°$, or 64 students $\implies 80°$, or 72 students $\implies 90°$. Let's check $90°$: $3x + 6 = 90 \implies x = 28$, and Swimming $2x - 1 = 55°$: $135 + 80 + 90 + 55 = 360$! Wait—the prompt and TikZ are already fixed in this JSON object, so let's make sure the prompt and TikZ use $(3x + 6)°$ and $(2x - 1)°$! Let's fix the prompt and TikZ right now before outputting!)

[Back to Question] *[Total Marks: 8]*

17 **(a)** Calculates Male $= 20$ and Female $= 40$ **[B1]**

Bar for Male drawn correctly from 0 to 20 with shading **[B1]**

Bar for Female drawn correctly from 20 to 60 without shading **[B1]**

(b) Total Females $= 30 + 45 + 40 = 115$ **[M1]**

Total Members $= 70 + 70 + 60 = 200$ and calculates $\frac{115}{200} \times 100$ **[M1]**

57.5 **[A1]**

(c) $\frac{40}{70}$ (or $\frac{4}{7}$) seen **[M1]**

$\frac{40}{70} \times \frac{39}{69}$ **[M1]**

$\frac{52}{161}$ or 0.323 (to 3 sf) **[A1]**

[Back to Question] *[Total Marks: 9]*

18 **(a)** Total Laptops $= 50 + 60 + 70 = 180$ **[B1]**

Total Tablets (Jan + Feb) $= 40 + 60 = 100$ **[B1]**

$180 = 1.2 \times (100 + x)$ or $\frac{180}{1.2} = 100 + x$ **[M1]**

$x = 50$ **[A1]**

(b) Total items (Jan to Mar) $= 180 + 150 = 330$ **[B1]**

Total items (April) $= 330 \times \frac{1}{3} = 110$ **[M1]**

Laptops (April) $= \frac{6}{6+5} \times 110$ **[M1]**

60 **[A1]**

(c) Tablet revenue $= 40 \times 200 = 8000$ and Laptop revenue $= 50 \times 450 = 22500$ **[M1]**

$\frac{8000}{8000 + 22500} \times 100$ **[M1]**

26.2 (accept 26.22 to 26.23) **[A1]**

[Back to Question] *[Total Marks: 11]*